Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable distributed systems, performance-tuned backends, and simulation pipelines. Experienced in C++, Python, SQL, concurrency, fault-tolerant workflows, database optimization, automated testing, and production reliability across AWS and Azure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected fault-tolerant distributed workflows processing 5,000+ monthly jobs with explicit state transitions, retries, reconciliation checks, and failure isolation, sustaining 99%+ production reliability while tripling concurrent throughput.
- Reduced multi-tenant dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling access paths and tuning Azure SQL indexing, queries, and normalized storage design.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a concurrent processing engine using multithreading and fault-tolerant execution, benchmarking throughput across 1,000+ files per batch and sustaining stable performance under 10,000+ daily requests.
- Built scalable Flask services with normalized persistence and per-operation usage instrumentation, reducing feedback turnaround 60% while enabling data-driven reliability and cost improvements across high-volume concurrent production workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built deterministic simulation and benchmarking pipelines over 100,000+ trajectory records, enabling reproducible experiments, controlled failure analysis, systematic performance comparison, and high-confidence debugging across large-scale closed-loop evaluation scenarios.
- Profiled and tuned a performance-critical PyTorch model through structured benchmarking and reproducible experimentation, improving long-horizon planning 18% and closed-loop stability 15% across repeated controlled large-scale simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered distributed APIs and ETL pipelines using object-oriented patterns, AWS Glue, and S3, improving response times 30% and pipeline efficiency 68% across reliable high-volume cross-platform data workflows.
- Built nightly VM-based simulation and test infrastructure with Jenkins, PyTest, Mockito, and 20+ suites across three operating systems, accelerating failure debugging and increasing reliability for distributed production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed distributed microservices with WebSocket synchronization across eight business workflows, scaling to 2 million+ Azure SQL records while preserving modular service boundaries, transactional consistency, and low-latency responsiveness.
- Automated Docker-based builds, integration tests, deployments, and operational monitoring through Jenkins and Azure services, cutting release time 50% while improving production stability across multiple controlled staged environments.

Skills

Languages & Systems: C++, Python, Java, SQL, JavaScript, TypeScript, Bash, Linux, object-oriented programming, algorithms
Distributed Systems: scalability, fault tolerance, concurrency, multithreading, asynchronous orchestration, microservices, WebSockets, failure isolation, high availability
Data & Performance: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, data modeling, indexing, query optimization, benchmarking, profiling, latency tuning
Testing & Cloud: simulation, automated testing, unit testing, integration testing, debugging, AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD, Git, PyTest, Mockito

Projects

CPU-Scheduling Simulator | *C++, Simulation, Operating Systems, React* | Live
- Built a C++ simulator modeling FCFS, SJF, SRT, Round Robin, priority, and multilevel queue scheduling with repeatable workloads, reducing average turnaround 25% across 1,000+ process configurations.

Chat2SQL | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker* | Code
- Built a cross-platform workspace supporting 10+ relational and NoSQL storage systems, with schema-aware query generation, structured validation, optimized execution, and Docker-backed deployment across Java and TypeScript services.

Big-Brain | *PostgreSQL, FAISS, Pinecone, GCP, REST APIs* | Live
- Engineered a cloud data ingestion and indexing pipeline processing 50,000+ records with incremental updates, metadata persistence, and performance tuning, reducing retrieval latency 40% through optimized execution paths.